PhotoConversion Materials
BSc Projects 2026

We are an interdisciplinary team working at the interface of physics, chemistry, and materials science to unveil fundamental mechanisms of light-matter interaction for energy conversion and optical sensing. We are located in the Onderzoeksgebouw on the VU campus, at De Boelelaan 1100.

Andrea Baldi	Charusheela Ramanan	Sven Askes	Loreta A. Muscarella
Nanomaterials for Energy Applications	**Disordered Molecular Materials**	**Pulsed Photothermal Catalysis**	**Materials for Piezo-phototronics**

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Extending the possibilities of non-toxic metal-free perovskites: making thin films

Contacts: Dr. Susan Rigter (s.a.rigter@vu.nl) and Dr. Loreta Muscarella (l.a.muscarella@vu.nl),

Level: BSc Thesis (3 months)

Metal-free perovskites (MFPs) are a new and interesting class of materials that remain underexplored in the scientific community. MFPs are attracting increasing interest as sustainable alternatives to conventional lead-based perovskites, offering tunable optoelectronic properties without the toxicity concerns [1].

In our lab, we create MFPs using ball milling, a simple but powerful technique that grinds together the starting powders and makes the targeted MFP. Compared to conventional solvent-based techniques, ball milling is a lot easier and more versatile, perfect for making new MFP materials.

However, the resulting material is a powder, while for many characterization techniques and applications, it would be preferable to have thin film samples: thin, uniform layers of material, supported by, for example, a flat piece of quartz. For other perovskite systems, we've found that even when the starting materials won't dissolve, the ball-milled products sometimes will, allowing us to make thin films via techniques such as spin coating. We believe our new MFPs could behave similarly, but we have not yet found the right strategy to achieve this.

Obtaining thin films will allow us to use many advanced characterization techniques, such as multimode AFM based techniques including piezo-electric force microscopy, to investigate this new class of materials on a fundamental level.

Tasks

The aim of this project is to find a method to transform ball milled metal-free powders into thin films on a substrate. Possible tasks include:

- Ball milling metal-free perovskite powders
- Finding a solvent from which the powder can be used in solution processing, such as spin coating or slot-die coating.
- Alternatively, investigating whether a thin film can be deposited via evaporation methods
- Extending this method to different compositions of metal-free perovskites made in our labs
- Using several techniques to characterize the resulting thin films and investigate how the powder and thin film materials compare to each other, such as XRD, UVvis, FTIR, optical microscopy, and profilometry.

References

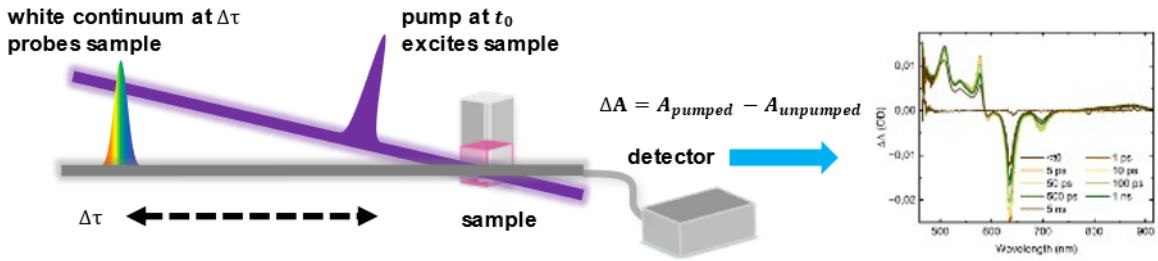
[1] Muscarella, L. A., Bravetti, G., & Milić, J. V. (2025). The emergence of metal-free molecular perovskites: challenges and opportunities. *Materials Horizons*.

Building and programming part of a femtosecond transient absorption setup

Contact: Vladimir Grigorescu (v.grigorescu@vu.nl), Charusheela Ramanan (c.ramanan@vu.nl)

Level: BSc Thesis (3 months)

Transient absorption spectroscopy is a nonlinear technique that is frequently employed to investigate the evolution of the photoexcited states of a material on timescales usually ranging from femto- to nanoseconds. The usual type of experiment that is performed is known as a "pump-probe" measurement, where a sample is first excited by a laser pulse (the "pump") and, after a known time delay, investigated with a second pulse (the "probe"). The difference in absorption between the excited and unexcited states yields a difference in absorption which evolves over time and can be analysed to reveal information about the photophysical processes taking place.



This experimental technique is highly relevant to many fields which require characterization of materials or systems that undergo ultrafast processes observable in the UV, visible or NIR regions of the spectrum, such as photocatalysis, plasmonics, photovoltaics, polaritonics, photonic computing etc.

The aim of this project is to complete the building and programming of a femtosecond transient absorption setup, with particular focus on the detection component. The main steps include familiarization with the equipment, planning out its placement on the optical table for optimal use of space, expanding on the existing software (python) and synchronization of the detector with other components, testing on known standards, standardizing the data output for compatibility with the data processing software and application to novel materials.

Depending on the speed of progress on the project, it can be restricted to the programming of the detector or can be expanded with development of an optimized measurement procedure dependent on experimental conditions.